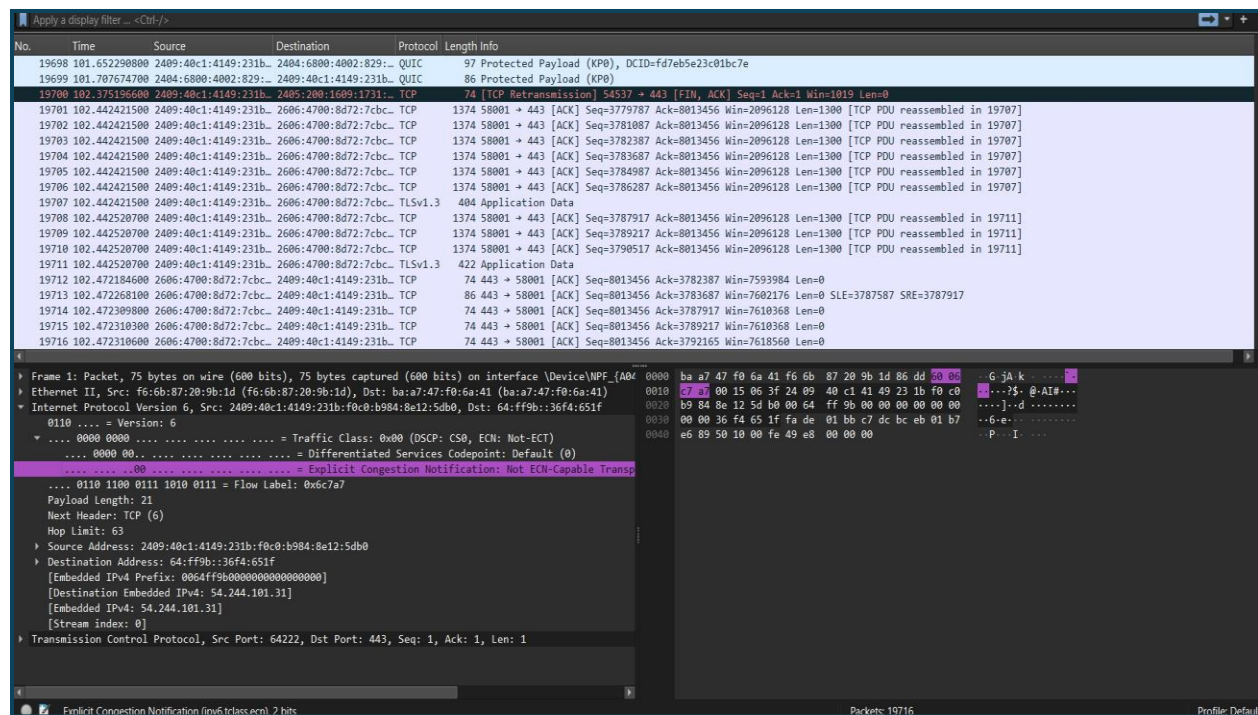


Day 1: Introduction to Networking and the OSI Model Theory :What is Networking? Overview of the OSI Model: Layers and their functions .Practical :Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers. Task: Create a diagram of the OSI Model with examples for each layer.

Tasks:-

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Ans. :-



Protocols observed:

1. QUIC
2. TCP (Transmission Control Protocol)
3. TLSv1.3 (Transport Layer Security)
4. IPv6 (Internet Protocol Version 6)

5. Ethernet II

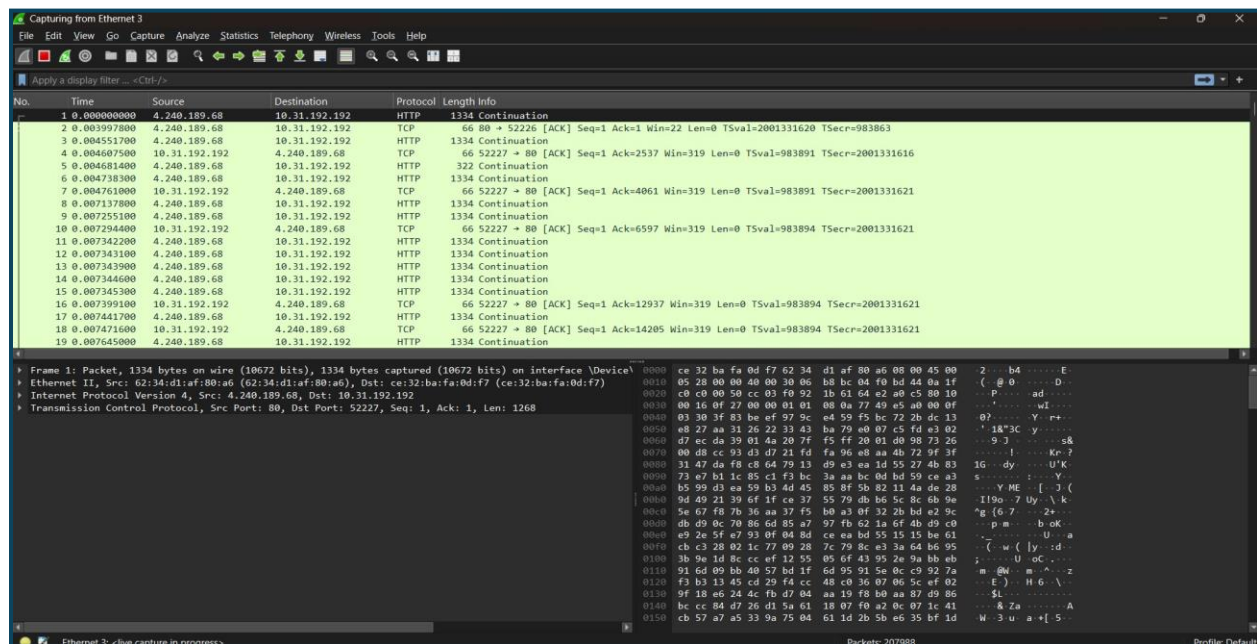
Description of the protocols:

- QUIC: A modern transport protocol used by HTTP/3 to provide secure and faster web communication.
- TCP: A reliable, connection-oriented protocol used for transmitting data between devices.
- TLSv1.3: Encrypts data exchanged between the web browser and the server to ensure secure communication.
- IPv6: The network-layer protocol used to route packets across the Internet.

2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

Ans.:-



Packet No.	Protocol	OSI Layer
Packet 1	HTTP	Layer 7 – Application
Packet 2	TCP	Layer 4 – Transport
Packet 1 (Packet Details)	IPv4	Layer 3 – Network

Brief explanation:

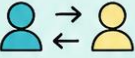
- HTTP – Application Layer (Layer 7): Used to transfer web pages and web content.
- TCP – Transport Layer (Layer 4): Provides reliable communication between devices.
- IPv4 – Network Layer (Layer 3): Responsible for addressing and routing packets across networks.

3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

Ans.:-

OSI MODEL

7 Layers of Networking

LAYER NO.	LAYER NAME	FUNCTION	REAL-WORLD EXAMPLE
7	Application Layer 	Provides network services directly to users.	When you use apps like WhatsApp, Chrome, Gmail, or YouTube.    
6	Presentation Layer 	Handles data formatting, encryption, compression and decryption.	When Instagram shows an image or a video in a nice format.  
5	Session Layer 	Manages sessions and connections between devices.	When you are on a video call on Zoom or Google Meet.  
4	Transport Layer 	Ensures complete data delivery with error handling and flow control.	When you send a file on WhatsApp, TCP ensures it reaches correctly.  
3	Network Layer 	Handles logical addressing and routing of data.	When your data travels from your home router to a server in another city.  
2	Data Link Layer 	Provides error detection and node-to-node data transfer.	When your device communicates with the Wi-Fi router (MAC address).  
1	Physical Layer 	Transmits raw bits over the physical medium (cables, signals, etc.).	When data is sent as electrical signals through Ethernet cable or Wi-Fi signals in the air.  



The OSI Model helps us understand how data travels from your device to the internet and back in 7 important steps!



4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

Ans.:-

WhatsApp Message Through the OSI Model :-

1. **Application Layer (Layer 7):**

I type a message in WhatsApp and press **Send**. The app prepares the message and sends it to the network.

2. **Presentation Layer (Layer 6):**

The message is encrypted and formatted so that only the intended receiver can read it securely.

3. **Transport Layer (Layer 4):**

The message is divided into small packets. TCP (or similar transport mechanisms) ensures the packets are delivered reliably and in the correct order.

4. **Network Layer (Layer 3):**

Each packet is given IP addresses and routed through the internet from my phone to WhatsApp's servers and then to the receiver's device.

5. **Data Link Layer (Layer 2):**

The packets are converted into frames and sent over the local Wi-Fi or mobile network using MAC addresses.

6. **Physical Layer (Layer 1):**

The frames are transmitted as electrical signals (Ethernet) or radio waves (Wi-Fi/mobile data) to the next network device.